

Arithmophysics

Geometry, quantum theory, the primes, and the border,
with the identities that cross them

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Abstract

Arithmophysics is the study of the primes through the objects of physics: oscillation, spectrum, confinement, with the dictionary between the two kept exact. This paper has four chapters, one for each strand the programme now has, and each chapter states only what is proved or computed in that strand: the geometry (the explicit formula as a sum of oscillators, the light cone, the horn, the fold, the curve), the quantum theory (the two Hamiltonians that exist, the Möbius state, the inner product that does not yet exist), the primes (the running proof's target and what has been proved around it), and the border (the Gödel theorem, the potential set, the conjecture). The fifth chapter is the point of the paper: seven crossings, each an exact identity or a proved equivalence between two strands, with its proof or citation, ending at the single object none of the four strands supplies. Every physical word in this paper names a theorem, a computation, or a labelled conjecture. Nothing here proves the Riemann Hypothesis.

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Status conventions and the rule of the word

A **Theorem**, **Proposition** or **Lemma** without a citation is proved here or in the cited document of the programme, where the proof is complete; with a citation to the literature it is taken from there with its hypotheses. A **Computation** carries its file in the repository. A **Crossing** is an identity or a proved equivalence joining two strands. A **Conjecture** is labelled and never used.

The rule of the word: in this paper “oscillator”, “Hamiltonian”, “horizon”, “light cone”, “trace”, “inner product”, “witness” each name a specific mathematical object defined where the word first appears. The physics is a dictionary, not an argument.

1 Geometry: the shape of the explicit formula

1.1 The sum of oscillators

For f real with compact support, $g = f \star \tilde{f}$, $\hat{g} = |\hat{f}|^2$, Weil’s explicit formula in the normalization of [8] reads

$$\sum_{\rho} \hat{g}\left(\frac{\rho - \frac{1}{2}}{i}\right) = W_{\infty}(g) + \hat{g}\left(\frac{i}{2}\right) + \hat{g}\left(-\frac{i}{2}\right) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (g(\log n) + g(-\log n)). \quad (1)$$

Read in $u = \log x$: the zero $\rho = \frac{1}{2} + i\gamma$ contributes the oscillator $e^{i\gamma u}$ to the prime side, and the prime power n contributes a point mass of weight $\Lambda(n)/\sqrt{n}$ at $u = \log n$ to the zero side. Each side is the spectrum of the other. In this paper “oscillator” means a term $e^{i\gamma u}$ of the left side of (1) and nothing else.

1.2 The light cone

For f supported in $[-a, a]$ only the prime powers with $\log n < 2a$ enter (1). This is the cone of [7, §12]: half-width a as time, $\log n$ as space, the edge $\log n = 2a$, the entry of n at $a_n = \frac{1}{2} \log n$. The odd floor $\lambda(a)$ is the infimum of the form over odd unit f supported in $[-a, a]$; it is attained and non-increasing [8, Lemma 1.4].

Theorem 1.1 (Weil [17]; Yoshida [18]). $\text{RH} \iff \lambda(a) \geq 0$ for every $a > 0$.

Theorem 1.2 (Zhu [19, Thm. 6.2]). $8.206 \times 10^{-15} \leq \lambda(0.8) \leq 2.347 \times 10^{-14}$, *certified*; so $\lambda > 0$ on $(0, 0.8]$.

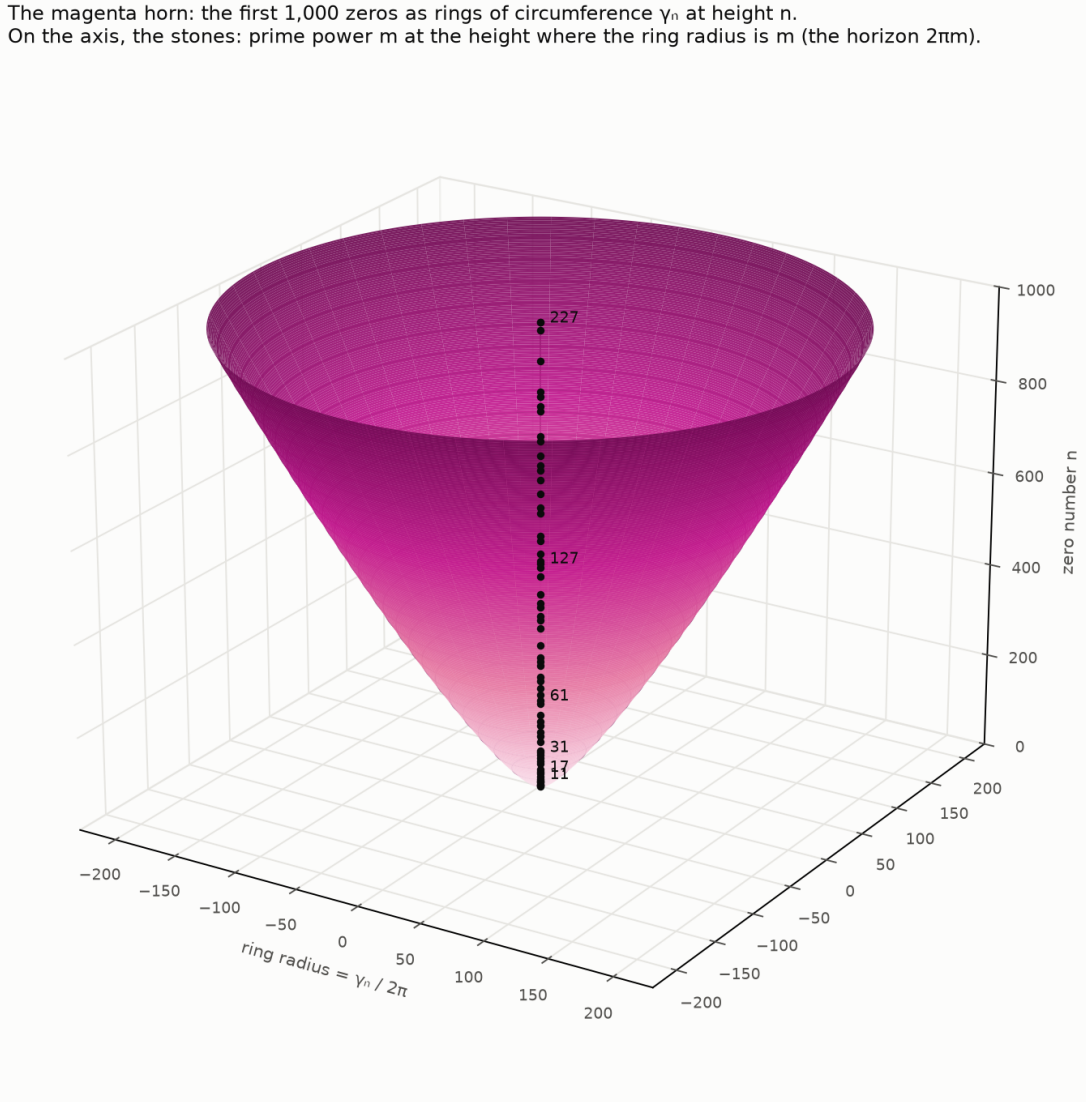
The horizon is $T^*(a) = 2\pi e^{2a}$. It is the height at which the density of zeros, $(2\pi)^{-1} \log(t/2\pi)$ by the Riemann–von Mangoldt law, equals the Cartwright bound a/π on the zero density of a function of exponential type a [8, §1]. This is an uncertainty relation between the two sides of (1): a window of half-width a on the prime line resolves the zeros only to the density a/π , that is, only below the horizon; and a function that would resolve them further has support beyond a and sees further primes.

Computation 1.3 (the floor; [7, 8]). From the files `floor_grid_K40.csv` and `floor_grid_arb.csv`: $\lambda(0.3) = 0.2226$, $\lambda(0.549) = 1.49 \times 10^{-5}$, $\lambda(1.2) = 7.2 \times 10^{-37}$, and $\lambda(1.505) \approx 10^{-93}$ by certified enclosure. The floor is positive at every support computed; the horizon at $a = 1.5$ is $T^* = 126$. Prime 2 enters at $a = 0.3466$ and is needed by $a = 0.3716$.

1.3 The horn

Draw the n -th zero as a circle of circumference γ_n at height n [7, §10]. Ring n has radius $r_n = \gamma_n/2\pi$, and the surface of revolution has profile the inverse of the counting function: $n = r(\log r - 1) + \frac{7}{8} + S$, $S = O(\log \gamma_n)$.

Computation 1.4 ([13]; `rh_horn.py`, 6,000 zeros). Growth per ring 0.2239 against $1/\log r = 0.2238$ (first 1,000 rings); rings minus law in $[-0.35, 1.36]$; Riemann–Siegel phase advance 1.00003π per ring; 85.3 percent of Gram intervals hold exactly one ring. The slope $1/\log r$ falls from 1.23 at ring 1 to 0.14 at ring 6,000: a horn, not a cone.



The horn drawn: the first 1,000 zeros as rings, with the stones on its axis, prime power m at the height where the ring radius is m (Crossing 1; file `MAGENTA_HORN.png`, script `rh_magenta_horn.py`).

The horn is hollow by construction, and only zeros on the line can be drawn as rings while the counting law counts every zero: for the Davenport–Heilbronn function the ring count drops by 2 at each off-line pair [7, Fig. 26C].

1.4 The fold

The functional equation is the reflection $s \mapsto 1 - \bar{s}$, whose fixed set is the line $\operatorname{Re} s = \frac{1}{2}$ [7, §8]. The exponent dictionary [1, Thm. 2.10]: a bound $M(x) \ll x^{1/2+\theta+\varepsilon}$ confines every zero to the shell $|\operatorname{Re} s - \frac{1}{2}| \leq \theta$, symmetric by the fold, hollow because the bound says nothing about the inside; what is known unconditionally is $\theta = \frac{1}{2} - o(1)$, the whole strip less a sliver; the Hypothesis is that the shell has thickness zero.

1.5 The curve

For a curve $X/\mathbb{F}_q$ of genus g with Frobenius eigenvalues $\alpha_i = \sqrt{q} e^{i\theta_i}$, the same construction gives the cone at support D (closed points of degree $\leq 2D$ inside, half-width $a = D \log q$), the Toeplitz form $T_{2D+1} = [\hat{\nu}_X(j-k)]$ with $\hat{\nu}_X(m) = q^{m/2} + q^{-m/2} - N_m q^{-m/2}$, and the horn with rings at radii $(\theta_i/2\pi + k)/\log q$.

Theorem 1.5 ([11, §2]). *T_{2D+1} is positive definite for $D < g$ and singular for $D \geq g$, with null vector the coefficients of $\prod_i (w - \bar{z}_i)$, $z_i = \alpha_i/\sqrt{q}$; the potential set is $\{\nu_X\}$ from $D = g$ on; the even sector closes at $D = g$, the odd at $g + 1$; and $T_{2g+1} \succeq 0$ if and only if every $|z_i| = 1$. The horn of the curve is an exact cone of slope $1/(2g \log q)$, because the zeros are periodic in height.*

The cone over a curve closes at $a = g \log q$; the cone over the integers never closes (Crossing 6).

2 Quantum theory: the operators

2.1 The stone-side Hamiltonian

On $\ell^2(\mathbb{N}) = \bigotimes_p \ell^2(\mathbb{N}_0)$, with $S_p e_n = e_{pn}$ and $H_{\text{arith}} e_n = (\log n) e_n$ [14, §2],

$$[H_{\text{arith}}, S_p] = (\log p) S_p, \quad \operatorname{tr} e^{-s H_{\text{arith}}} = \zeta(s) \quad (\operatorname{Re} s > 1).$$

This is Julia's primon gas and the Bost–Connes system [20]. Its eigenvalues are the positions of the stones on the prime line; its partition function is ζ where ζ has no zeros. “Hamiltonian” here means a self-adjoint operator whose eigenvalues are the positions of one side of (1).

2.2 The rule-side Hamiltonian

On $L^2((0, \infty), dx)$ the dilation $(U_t f)(x) = e^{t/2} f(e^t x)$ is unitary [2, Part VI], $U_t = e^{it H_{\text{dil}}}$ with

$$H_{\text{dil}} = -i \left(x \frac{d}{dx} + \frac{1}{2} \right) = \frac{xp + px}{2}, \quad H_{\text{dil}} x^{\rho-1} = -i \left(\rho - \frac{1}{2} \right) x^{\rho-1} = [\gamma - i(\beta - \frac{1}{2})] x^{\rho-1}$$

for $\rho = \beta + i\gamma$. This is the operator of Berry and Keating [21]; the mode of a zero has a real eigenvalue exactly when the zero is on the line; and the modes are not in L^2 . The operator has continuous spectrum filling $\mathbb{R}$ and selects no zero [22]. In the coordinate $u = \log x$, $(\mathcal{T}f)(u) = e^{u/2} f(e^u)$ conjugates it to $-i d/du$: the oscillators of Section 1 are its generalized eigenfunctions.

2.3 The Möbius state

Proposition 2.1 ([1, Prop. 3.2]). *Let $m(u) = e^{-u/2} M(e^u) = (\mathcal{T}M(x)/x)(u)$. Then (i) $I_M(e^U) = \int_0^U |m|^2$; (ii) $\int_0^\infty m(u) e^{-(s-1/2)u} du = 1/(s\zeta(s))$ for $\operatorname{Re} s > 1$; (iii) $m \notin L^2(0, \infty)$, unconditionally.*

The state whose power is the running proof's energy has spectrum $1/(s\zeta(s))$ on the line, resonances at the zeros, and infinite total power.

2.4 The inner product

Theorem 2.2 (Connes [23]). *RH is equivalent to the positivity of the Weil distribution: to the existence of an inner product on the test functions in which (1) is the trace of the action of the idele class group, the zeros appearing as an absorption spectrum.*

The floor $\lambda(a)$ is this positivity at support a . The operator whose self-adjointness in that inner product would put the zeros on the line is not constructed; the six constraints it must satisfy are [1, §4].

2.5 The curve, where every piece exists

Theorem 2.3 (Weil [26]; trace formula [27, Ch. VI]). $N_m = 1 + q^m - \text{tr}(F^m | H^1(X))$, $|\alpha_i| = \sqrt{q}$. With $U = F/\sqrt{q}$ and $H_X = -i \log U$, $N_m = 1 + q^m - q^{m/2} \text{tr} e^{imH_X}$, and RH_X is the unitarity of U , supplied by the positivity of the Rosati involution (genus one: Hasse's $\deg(m - nF) = m^2 - a_1 mn + qn^2 \geq 0$) and the Hodge index theorem on $X \times X$.

Katz and Sarnak [28] proved the random-matrix statistics of the θ_i over families; for ζ the same statistics are Montgomery's theorem in its restricted range and Odlyzko's computation [29, 30].

3 The primes: the record

3.1 The target

$M(x) = \sum_{n \leq x} \mu(n)$, $I_M(N) = \int_1^N M(t)^2 t^{-2} dt$. The running proof's single target is

$$I_M(X) \leq C_\varepsilon X^\varepsilon \quad (X \geq 2) \text{ for every } \varepsilon > 0. \quad (\text{H})$$

Theorem 3.1 ([3, §11]; Littlewood [31, §14.25]). $(\text{H}) \iff \text{RH}$. *The forward direction is the Mellin step: (H) gives absolute convergence of $\int_1^\infty M(x)x^{-s-1}dx$ on $\text{Re } s > \frac{1}{2}$, hence holomorphy of $1/\zeta$ there, hence no zeros there, and the fold removes the other side.*

3.2 What is proved around it

Theorem 3.2 ([2, Part XVII]). $I_M(N) = \sum_{u \leq N} w_u(1/u - 1/\min(v_u, N))$ over the levels of $|M|$, and $\mathcal{E}_*(N) \leq I_M(N) \leq 2 \log N + \mathcal{E}_*(N)$.

Theorem 3.3 ([4, Thm. 3.1, §4, §7]). $H_{\text{record}} \iff \text{RH}$. *The debit identity $\Delta_N = E(u_0) + 2M(m)h(u_0) + E(W) - 2\langle a, W \rangle$ is exact; its two positive terms are each $\geq (1 - o(1))m/(24 \log^2 m)$; the signed cross term carries the cancellation; the cofactor-flow identity $E(x - U_p x) = (1 + \frac{1}{p})R(L) - \frac{2}{p}M(L)m_1(L)$ holds with $|m_1| \leq 1$.*

Theorem 3.4 ([5], checked in [6]). *Every target of the lifetime census is equivalent to RH; every proved credit is below the allowance $N^{1/2+\varepsilon}$; the paid insertion range is the Legendre range $2^{\pi(Y)} \leq \sqrt{N}$, beyond which the alternating cofactor sum meets the parity obstruction.*

Theorem 3.5 ([14]). *For $a > 1$ the Möbius coefficient state $m_a = \sum \mu(d)d^{-a/2}e_d$ has $\|m_a\|^2 = \zeta(a)/\zeta(2a)$; a global bilateral all-prime unitary exists; the cumulative synthesis map is unbounded; its Euler update is $q_\alpha((I - r_p S_p)c) = (1 + r_p^2)q_\alpha(c) - 2r_p \text{Re}\langle W_\alpha c, D_p W_\alpha c \rangle$; the shell-ledger gate along a determining sequence is equivalent to RH.*

Every quantity proved in this strand is one the target already tolerates; the subpower exponent is the only non-finitary object in it, and it sits at infinity.

4 The border: the logic

Theorem 4.1 ([10, Thms. 4.1, 5.1, 6.1]; [32]). *RH is ZFC-equivalent to a Π_1^0 sentence ρ . Adding every true Σ_1^0 sentence to any theory containing Robinson arithmetic adds no theorem. $\text{PA} \vdash \rho \leftrightarrow \text{Con}(\text{Q} + \rho)$ and $\text{ZFC} \vdash \rho \leftrightarrow \text{Con}(\text{PA} + \rho)$. A refutation of ρ has a witness provable in Q; non-refutability implies truth. Any true principle establishing ρ over PA is unprovable in PA, not equivalent to a Σ_1^0 sentence, and proves $\text{Con}(\text{Q} + \rho)$.*

Proposition 4.2 ([9, Props. 2.2–2.3]; [1, Prop. 2.11]). *The potential set $\mathcal{P}(a)$ of configurations the primes inside the cone cannot distinguish from the zeros is non-increasing, nonempty iff the form is nonnegative at support a , and $\bigcap_a \mathcal{P}(a) = \{\nu_\zeta\}$ iff RH. Under RH the form has no null vector at any finite support: the collapse happens only at $a = \infty$.*

On a curve the pattern has a last stone (degree $2g$), RH_X for one curve is a finite check, and the positivity that closes it, Weil’s, is a theorem; the Gödel constraints bite only where the pattern has no last stone. *The Grammar of Things* [16] says the same in the language of a circle of stones: the records and the rule are one object described twice, and the distance between them is a border, not a distance that accumulation crosses.

Conjecture 4.3 (Michaels; [10, §10]). ρ is not provable in ZFC. If so, RH is true, every instance is provable, ρ is independent, and the inner product of Section 2 is a new universal axiom.

Labelled a conjecture; used nowhere.

5 The crossings

Each crossing joins two strands by an identity or a proved equivalence. The proofs are one line or a citation; the point is that they are exact.

Crossing 1 (geometry $\times$ primes: the stones sit in the horn). The prime power m enters the cone at $a_m = \frac{1}{2} \log m$, where $T^*(a_m) = 2\pi m$: in the horn’s unit, ring radius m . The stone m sits on the horn’s axis at radius m and has resolved $N(2\pi m) = m(\log m - 1) + \frac{7}{8} + O(\log m)$ rings.

Proof. $T^*(a_m) = 2\pi e^{\log m} = 2\pi m$ and $r = T/2\pi$. The count is the Riemann–von Mangoldt law at $T = 2\pi m$. Computed for every prime power to 1,000 in `data/horn_stones.json`: $2 \mapsto 0$ (the horizon 4π is below the first zero), $3 \mapsto 1$, $7 \mapsto 8$, $11 \mapsto 16$, $97 \mapsto 348$, $997 \mapsto 5,888$. $\square$

Crossing 2 (geometry $\times$ quantum: rings are wavenumbers). The ring of the zero γ has radius $\gamma/2\pi$, the number of oscillations per e -fold of x of the oscillator $e^{i\gamma \log x}$; the horn’s slope $1/\log r$ is the mean spacing of the zeros in that unit; the Riemann–Siegel phase advances by π per ring.

Proof. The oscillator has period $2\pi/\gamma$ in $u = \log x$; $r = 1/\text{period}$. The mean gap at height T is $2\pi/\log(T/2\pi)$, which is $1/\log r$ in radius. $\theta(t) = \pi[r(\log r - 1) - \frac{1}{8}] + O(1/t)$ and $N = \theta/\pi + 1 + S$; Computation 1.4 measures 1.00003 π . $\square$

Crossing 3 (quantum $\times$ primes: the target is a spectral statement). (H) holds iff the Möbius state has subexponential accumulated power, $\int_0^U |m|^2 \leq C_\epsilon e^{\epsilon U}$; its transform is $1/(s\zeta(s))$; RH is the statement that every resonance of the state has a real frequency.

Proof. Proposition 2.1(i)–(ii) and Theorem 3.1. $\square$

Crossing 4 (geometry $\times$ quantum, on the curve: the cone closes when the matrix runs out). $N_m = 1 + q^m - q^{m/2} \operatorname{tr} e^{imH_X}$; the Toeplitz form of the cone at support D is $\sum_i |\hat{h}(\theta_i)|^2$ over the $2g$ eigenphases; it acquires a null vector exactly when $2D + 1 > 2g$, and the null vector's roots are the eigenphases. The horn of the curve is an exact cone because H_X has finitely many eigenvalues, periodic in height.

Proof. Theorems 2.3 and 1.5; the genus-one case by hand in [12]: $N_1 = 6$, $N_2 = 12$, $a_1 = -2$, the 3×3 form at $a = \log 3$ with eigenvalues $0, \frac{8}{3}, \frac{10}{3}$, null vector $(1, 2/\sqrt{3}, 1)$ with roots at $\cos \theta = -1/\sqrt{3}$. $\square$

Crossing 5 (primes $\times$ quantum: one signed cross term, in two normalizations). The Euler update of Theorem 3.5 and the cofactor-flow identity of Theorem 3.3 have the same shape: a positive diagonal $(1 + r_p^2)q$, respectively $(1 + \frac{1}{p})R$, and one signed cross term, $-2r_p \operatorname{Re}\langle Wc, D_p Wc \rangle$, respectively $-\frac{2}{p}M(L)m_1(L)$. In both, positivity fixes the total to be a square and leaves the sign of the cross term open; in both, that sign is the whole question, and in the census it is the parity obstruction.

Proof. Read the two identities side by side; the second is [4, M44], the first [14, §3]. The parity statement is Theorem 3.4. $\square$

Crossing 6 (border $\times$ all: the edge, and what kind of thing each object is). The following are equivalent, each equivalence proved: RH; (H); H_{record} ; $\lambda(a) \geq 0$ for all a ; $\bigcap_a \mathcal{P}(a) = \{\nu_\zeta\}$; the census residual bound; over ZFC, $\operatorname{Con}(\text{PA} + \rho)$. Under RH no finite support decides any of them. The inner product is a universal statement; a negative floor at one finite support, or one off-line zero, is a witness with a finite certificate.

Proof. [1, Thm. 2.10, Prop. 2.11]; the witness clause is Theorem 4.1 with Theorem 1.1: $\lambda(a) < 0$ is certifiable by interval arithmetic exactly as Theorem 1.2 certifies $\lambda(0.8) > 0$. $\square$

Remark 5.1 (the forced case). The Davenport–Heilbronn function keeps the functional equation and drops the Euler product; its floor turns negative at a finite support, near $a \approx 2.1$, where the horizon reaches its off-line zero at height 85.7 [7, Rem. 12.9], and its horn loses two rings per pair. That is a witness for a cousin, and it says nothing about ζ , which is the Euler-product filter of [15]. The same distinction appeared in fluid mechanics in September 2026, when a forced Navier–Stokes blow-up with bounded energy was announced while the unforced question stayed open [33]: a witness for a forced cousin, the conserved energy blind to it, the universal statement untouched. The parallel is in the logical type of the statements, not in the equations.

Crossing 7 (border $\times$ curve: where the pattern is finite the axiom is a theorem). On a curve the stones end at degree $2g$, the cone closes at $a = g \log q$, positivity at the single support $D = g$ is RH_X , and Weil's positivity is a theorem of ZFC. Over the integers the stones do not end, the cone does not close, no support decides, and any principle that closes it proves $\operatorname{Con}(\mathbb{Q} + \rho)$.

Proof. Theorems 1.5, 2.3, 4.1; Proposition 4.2. $\square$

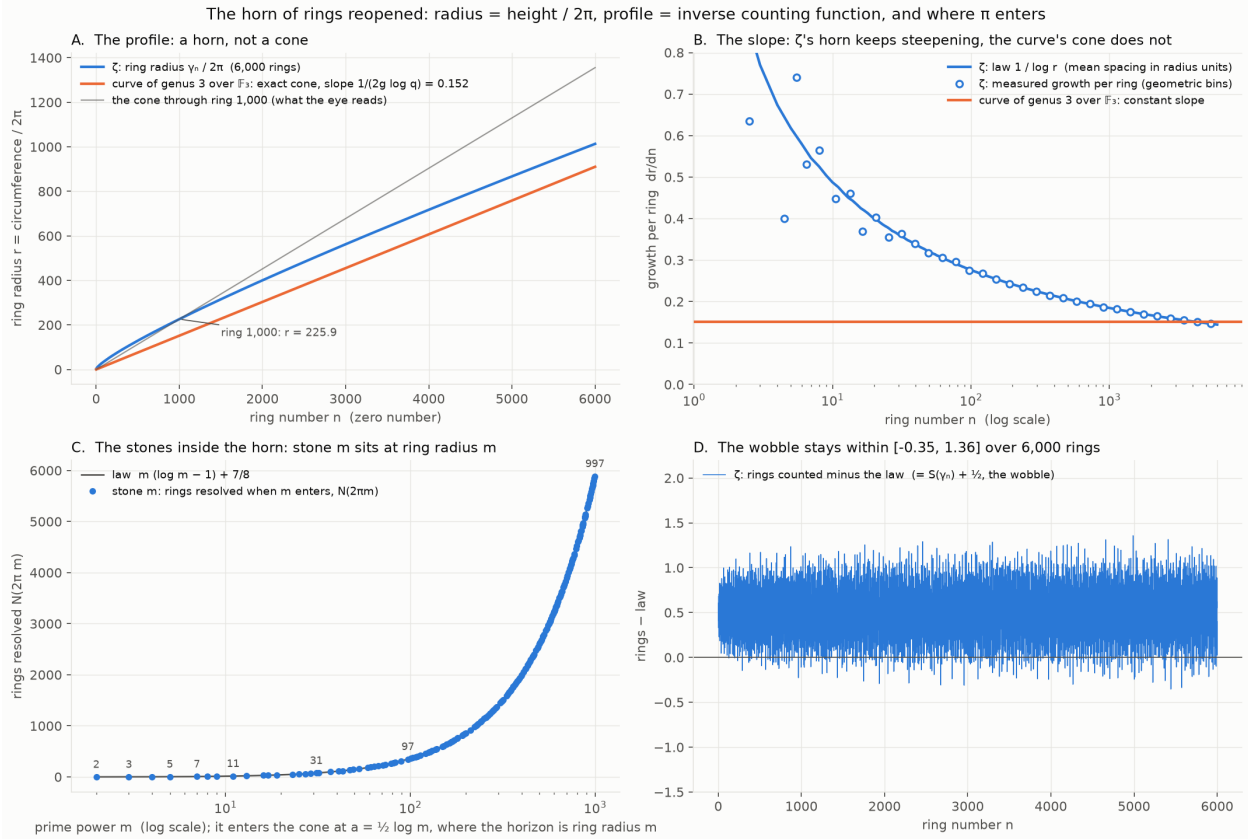
5.1 The one object none of the strands supplies

Geometry supplies the shape and shows where it closes. Quantum theory supplies two operators and names the third. The primes supply the record and locate its end at infinity. The border says what kind of statement the missing object's positivity must be. What none of them supplies is the inner product for ζ : the space in which (1) is a trace, the positivity of which is the Hypothesis.

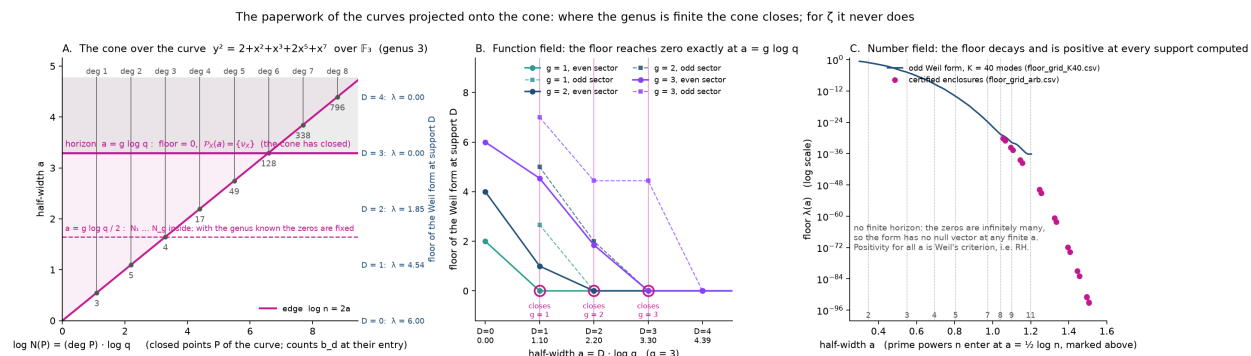
Its specification is the six constraints of [1, §4]: the trace with the prime sign handled, positivity equivalent to RH, infinite genus with the growth of $N(T)$, entries as orbit half-lengths, the axiom-from-above form, and the function-field template. Deninger’s foliated spaces [25] and the arithmetic site of Connes and Consani [24] are the candidates, each stalled at exactly the positivity. Arithmophysics, in the sense of this paper, is the study of that one missing object through the four that exist.

6 Status

statement	status	where
explicit formula; Weil’s criterion; Zhu’s certified floor	cited	§1
cone, horizon, floor decay; horn, phase per ring, Gram’s law	computed	Comp. 1.3, 1.4
cone over a curve closes at $g \log q$; finite-support criterion	proved	Thm. 1.5
two Hamiltonians; Berry–Keating identity; continuous	proved / cited	§2
spectrum		
Möbius state: power, transform, infinite power	proved	Prop. 2.1
RH $\iff$ positivity of the Weil distribution	cited	Thm. 2.2
curve: trace formula, unitarity, Katz–Sarnak	cited	Thm. 2.3
(H) $\iff$ RH; lifetime truncation; H_{record} ; census; Build v32	proved / cited	§3
the subpower premise (H)	open	
Gödel theorems; potential set; no null vector at finite a	proved	§4
Crossings 1–7	proved	§5
the inner product for ζ	not constructed	§5
the Michaels Conjecture	conjecture	Conj. 4.3
the Riemann Hypothesis	open	



The horn: profile and slope against the exact cone of a curve, the stones inside at ring radius m , and the wobble (THE_HORN.png).



The cone over a curve closing at $g \log q$; the number-field floor decaying and never reaching zero (FUNCTION_FIELD_CONE.png).

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